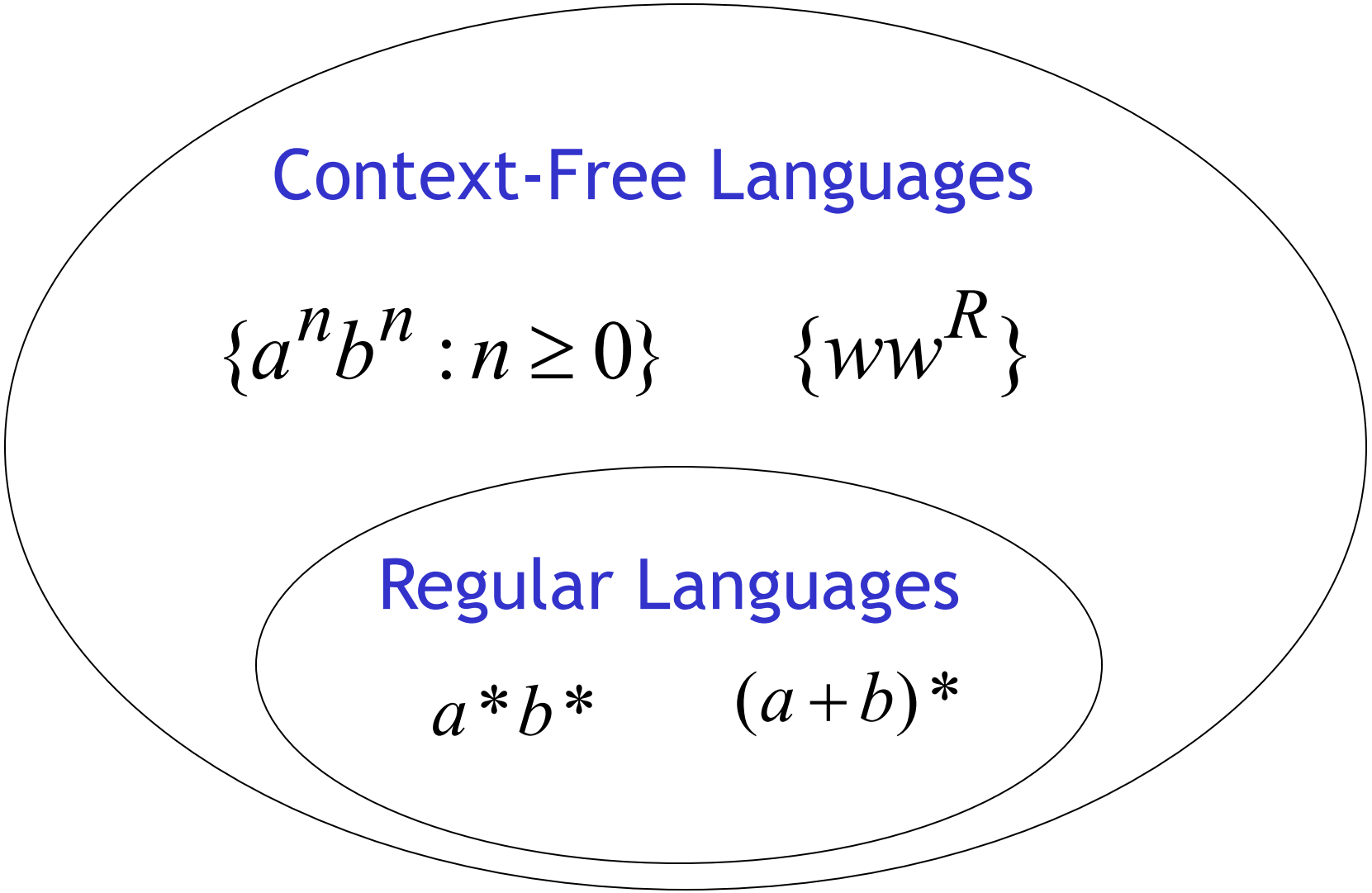
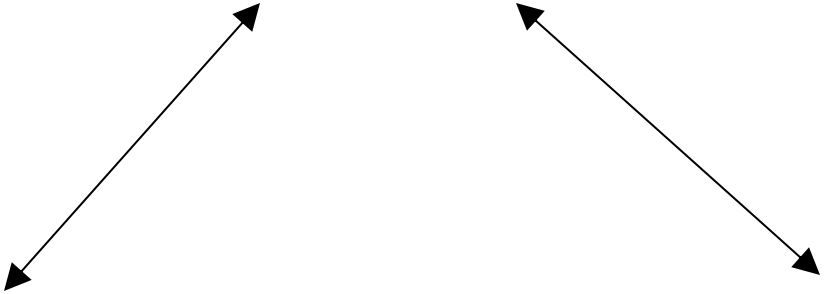
# Context-Free Languages

Original Slides by C. Busch, LSU. Modified by A. H. Özer, Marmara University.

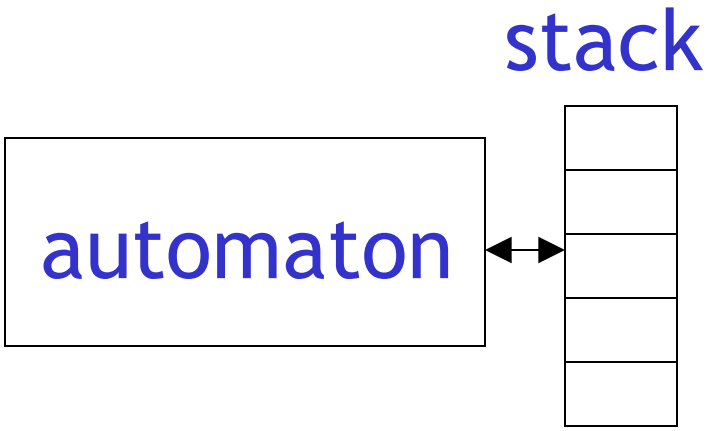

Context-Free Languages
$\{a^n b^n : n \geq 0\}$
$\{ww^R\}$
Regular Languages
$a*b*$
$(a+b)*$

# Context-Free Languages



Context-Free Grammars

Pushdown Automata



# Context-Free Grammars

# Grammars

Grammars express languages

Example: the English language grammar

$$\langle sentence \rangle \rightarrow \langle noun_phrase \rangle \ \langle predicate \rangle$$

$$\langle noun_phrase \rangle \rightarrow \langle article \rangle \ \langle noun \rangle$$

$$\langle predicate \rangle \rightarrow \langle verb \rangle$$

$$\langle article \rangle \rightarrow a$$

$$\langle article \rangle \rightarrow the$$

$$\langle noun \rangle \rightarrow cat$$

$$\langle noun \rangle \rightarrow dog$$

$$\langle verb \rangle \rightarrow runs$$

$$\langle verb \rangle \rightarrow sleeps$$

# Derivation of string "the dog sleeps":

$$\begin{aligned}
\langle sentence \rangle &\Rightarrow \langle noun_phrase \rangle \ \langle predicate \rangle \\
&\Rightarrow \langle noun_phrase \rangle \ \langle verb \rangle \\
&\Rightarrow \langle article \rangle \ \langle noun \rangle \ \langle verb \rangle \\
&\Rightarrow the \ \langle noun \rangle \ \langle verb \rangle \\
&\Rightarrow the \ \ dog \ \langle verb \rangle \\
&\Rightarrow the \ \ dog \ \ sleeps
\end{aligned}$$

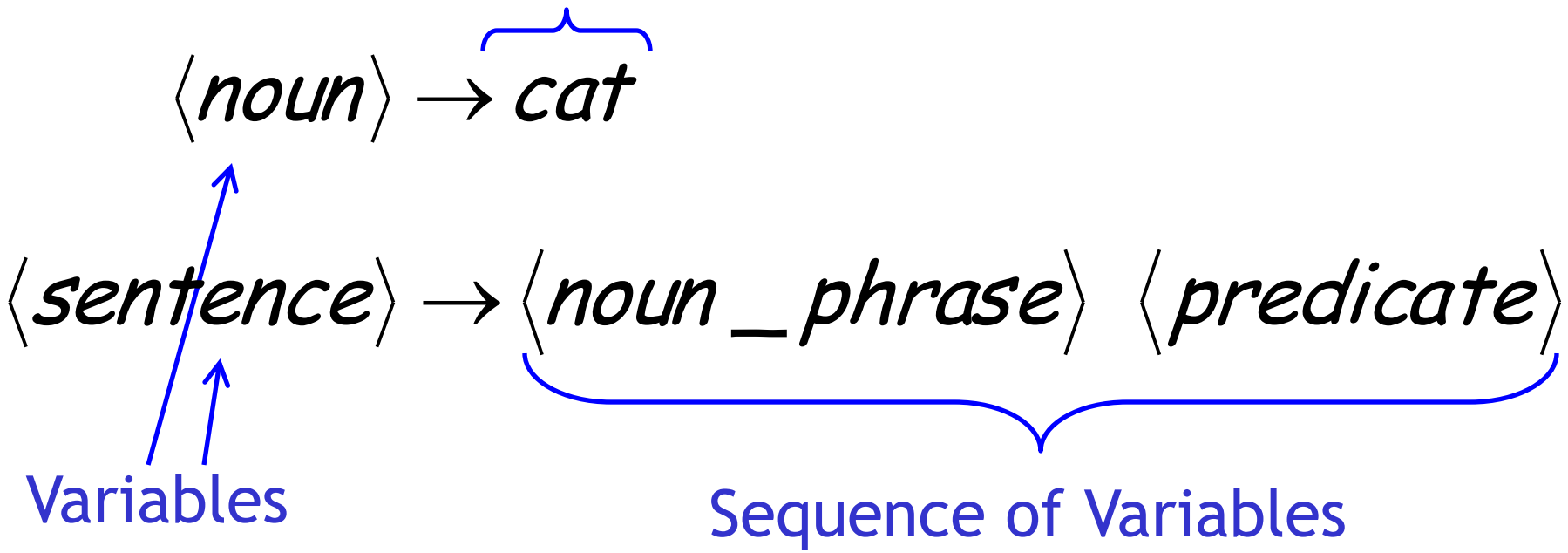
# Derivation of string “a cat runs”:

$$\begin{aligned}
\langle sentence \rangle &\Rightarrow \langle noun_phrase \rangle \ \langle predicate \rangle \\
&\Rightarrow \langle noun_phrase \rangle \ \langle verb \rangle \\
&\Rightarrow \langle article \rangle \ \langle noun \rangle \ \langle verb \rangle \\
&\Rightarrow a \ \langle noun \rangle \ \langle verb \rangle \\
&\Rightarrow a \ \ cat \ \langle verb \rangle \\
&\Rightarrow a \ \ cat \ \ runs
\end{aligned}$$

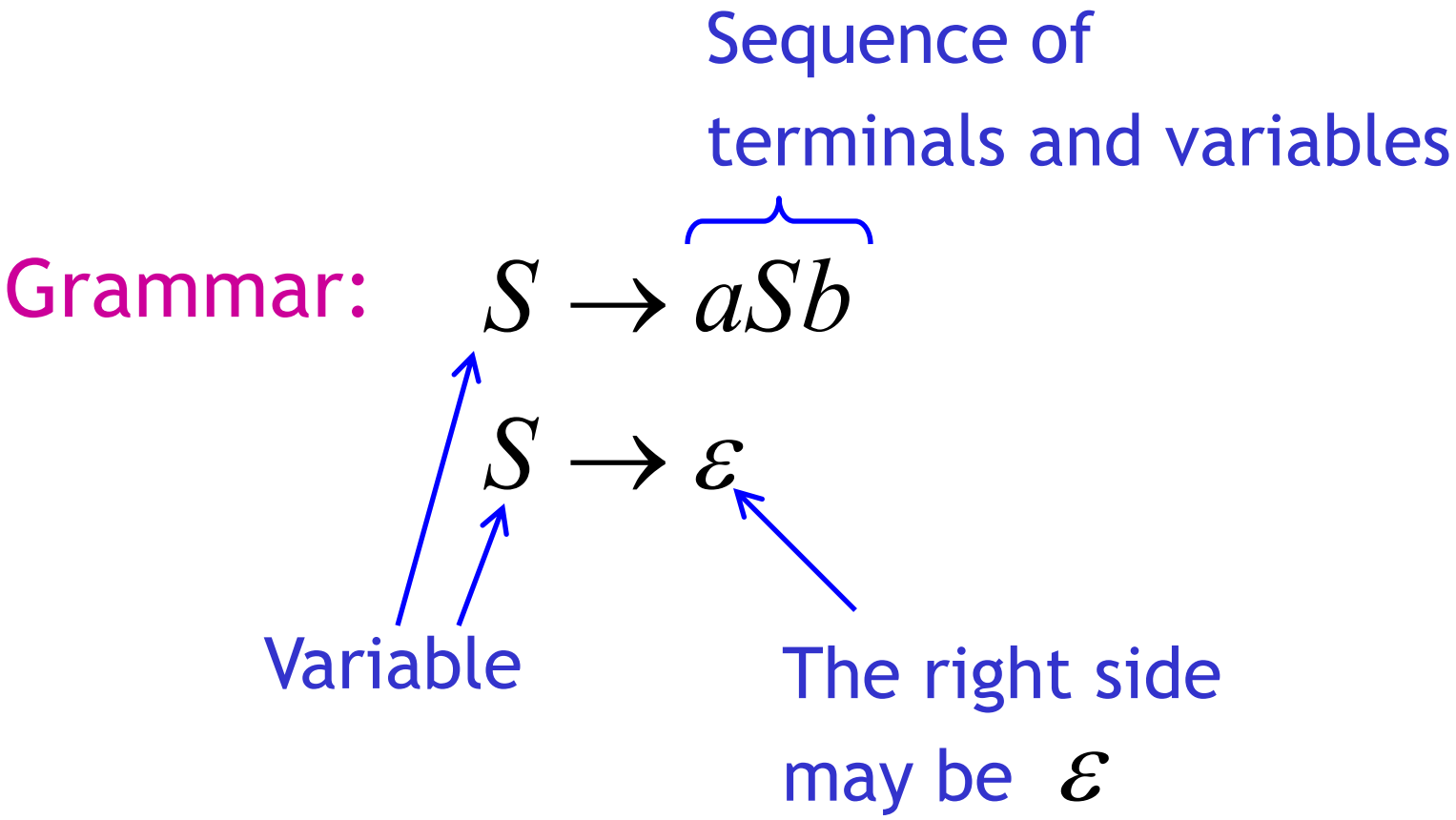
## Language of the grammar:

L = { “a cat runs”,
“a cat sleeps”,
“the cat runs”,
“the cat sleeps”,
“a dog runs”,
“a dog sleeps”,
“the dog runs”,
“the dog sleeps” }

# Productions



# Another Example



Grammar: $S \to aSb$

$S \to \varepsilon$

Derivation of string $ab$ :

$$S \Rightarrow aSb \Rightarrow ab$$

$S \to aSb$ $\qquad$ $S \to \varepsilon$

Grammar: $S \to aSb$

$S \to \varepsilon$

Derivation of string $aabb$ :

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aabb$$



$S \to aSb$ $\qquad$ $S \to \varepsilon$

Grammar: $S \to aSb$

$S \to \varepsilon$

Other derivations:

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaaSbbb \Rightarrow aaabbb$$

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaaSbbb$$

$$\Rightarrow aaaaSbbbb \Rightarrow aaaabbbb$$

Grammar: $S \to aSb$

$S \to \varepsilon$

Language of the grammar:

$$L = \{a^n b^n : n \geq 0\}$$

# A Convenient Notation

We write: $S \overset{*}{\Rightarrow} aaabbb$

for zero or more derivation steps

Instead of:

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaaSbbb \Rightarrow aaabbb$$

In general we write: $w_1 \overset{*}{\Rightarrow} w_n$

If: $w_1 \Rightarrow w_2 \Rightarrow w_3 \Rightarrow \cdots \Rightarrow w_n$

in zero or more derivation steps

Trivially: $w \overset{*}{\Rightarrow} w$

## Example Grammar

$$S \to aSb$$

$$S \to \varepsilon$$

## Possible Derivations

$$S \overset{*}{\Rightarrow} \varepsilon$$

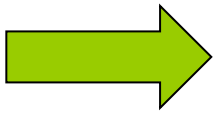
$$S \overset{*}{\Rightarrow} ab$$

$$S \overset{*}{\Rightarrow} aaabbb$$

$$S \overset{*}{\Rightarrow} aaSbb \overset{*}{\Rightarrow} aaaaaSbbbbb$$

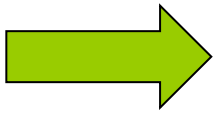
Another convenient notation:

$$S \to aSb$$
$$S \to \varepsilon$$



$$S \to aSb \mid \varepsilon$$

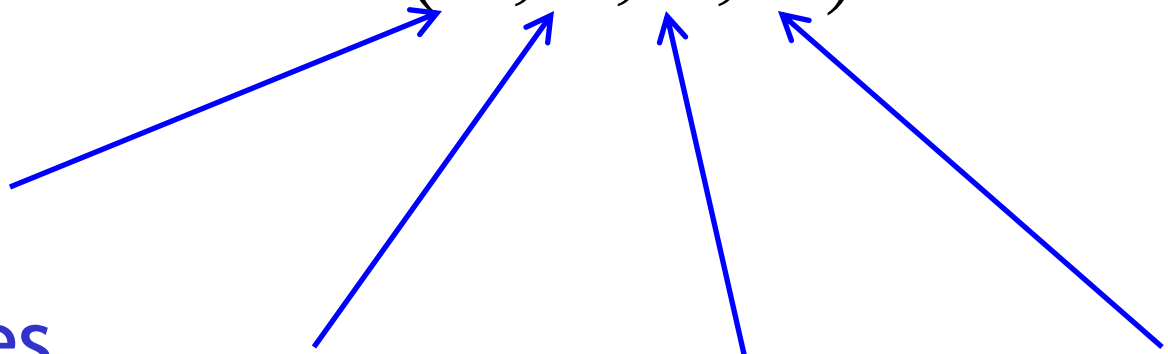
$$\langle article \rangle \to a$$
$$\langle article \rangle \to the$$



$$\langle article \rangle \to a \mid the$$

# Formal Definitions

Grammar: $G = (V, T, S, P)$



Set of variables

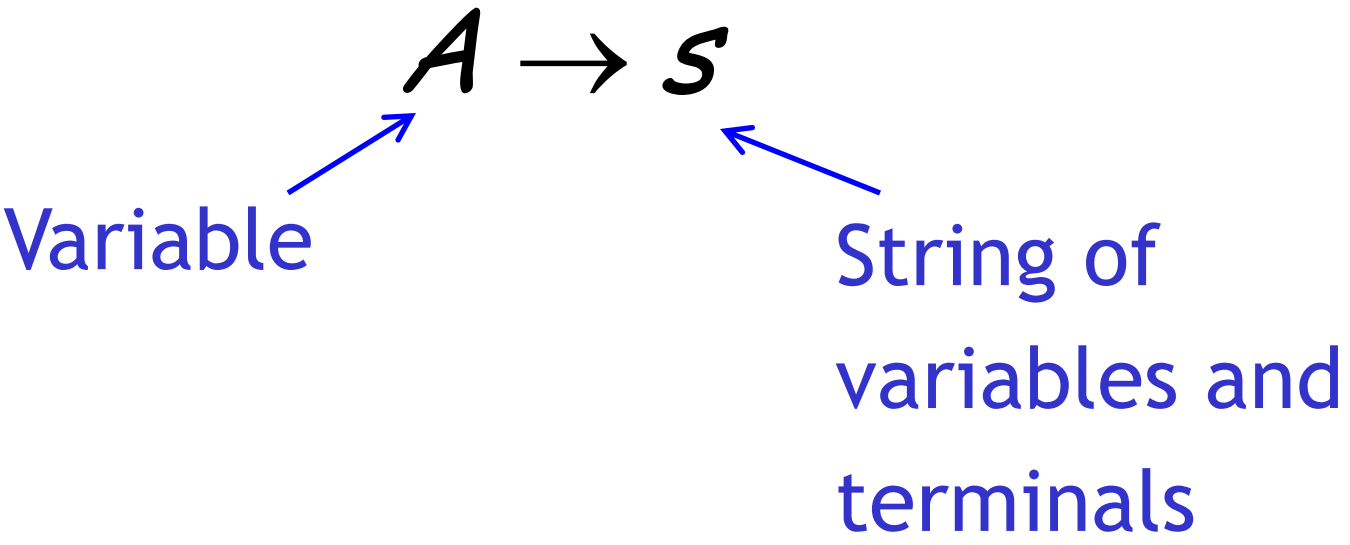
Set of terminal symbols

Start variable

Set of productions

# Context-Free Grammar: $G = (V, T, S, P)$

All productions in **P** are of the form



# Example of Context-Free Grammar

$$S \to aSb \mid \varepsilon$$

productions

$$P = \{S \to aSb,\ \ S \to \varepsilon\}$$

$$G = (V, T, S, P)$$

$V = \{S\}$

variables

$T = \{a, b\}$

terminals

start variable

# Language of a Grammar:

For a grammar $G$ with start variable $S$

$$L(G) = \{w : \quad S \overset{*}{\Rightarrow} w, \quad w \in T^*\}$$

String of terminals or $\varepsilon$

## Example:

Context-free grammar $G$ : $\boxed{S \to aSb \mid \varepsilon}$

$$L(G) = \{a^n b^n : \quad n \geq 0\}$$

Since, there is derivation

$$S \overset{*}{\Rightarrow} a^n b^n \quad \text{for any} \quad n \geq 0$$

## Context-Free Language definition:

A language $L$ is context-free

if there is a context-free grammar $G$

with $L = L(G)$

# Example:

$$L = \{a^n b^n : \quad n \geq 0\}$$

is a context-free language

since context-free grammar $G$:

$$\boxed{S \to aSb \mid \varepsilon}$$

generates $L(G) = L$

# Another Example

Context-free grammar $G$ :

$$S \to aSa \mid bSb \mid \varepsilon$$

Example derivations:

$$S \Rightarrow aSa \Rightarrow abSba \Rightarrow abba$$

$$S \Rightarrow aSa \Rightarrow abSba \Rightarrow abaSaba \Rightarrow abaaba$$

---

$$L(G) = \{ww^R : \quad w \in \{a,b\}^*\}$$

Palindromes of even length

# Another Example

Context-free grammar $G$ :

$$S \to aSb \mid SS \mid \varepsilon$$

Example derivations:

$$S \Rightarrow SS \Rightarrow aSbS \Rightarrow abS \Rightarrow ab$$

$$S \Rightarrow SS \Rightarrow aSbS \Rightarrow abS \Rightarrow abaSb \Rightarrow abab$$

---

$$L(G) = \{w \ : n_a(w) = n_b(w), \text{ and } n_a(v) \geq n_b(v) \text{ in any prefix } v\}$$

Describes matched parentheses:

() ((( ))) (( ))  $a = (, \quad b = )$

# Derivation Order
# and
# Derivation Trees

# Derivation Order

Consider the following example grammar with 5 productions:

| | | |
|---|---|---|
| 1. $S \to AB$ | 2. $A \to aaA$ | 4. $B \to Bb$ |
| | 3. $A \to \varepsilon$ | 5. $B \to \varepsilon$ |

1. $S \to AB$ 2. $A \to aaA$ 4. $B \to Bb$

3. $A \to \varepsilon$ 5. $B \to \varepsilon$

Leftmost derivation order of string *aab*:

$$S \overset{1}{\Rightarrow} AB \overset{2}{\Rightarrow} aaAB \overset{3}{\Rightarrow} aaB \overset{4}{\Rightarrow} aaBb \overset{5}{\Rightarrow} aab$$

At each step, we substitute the leftmost variable.

1. $S \to AB$ 2. $A \to aaA$ 4. $B \to Bb$

3. $A \to \varepsilon$ 5. $B \to \varepsilon$

Rightmost derivation order of string $aab$ :

$$S \overset{1}{\Rightarrow} AB \overset{4}{\Rightarrow} ABb \overset{5}{\Rightarrow} Ab \overset{2}{\Rightarrow} aaAb \overset{3}{\Rightarrow} aab$$

At each step, we substitute the rightmost variable.

1. $S \to AB$ 2. $A \to aaA$ 4. $B \to Bb$

3. $A \to \varepsilon$ 5. $B \to \varepsilon$

Leftmost derivation of $aab$ :

$$S \overset{1}{\Rightarrow} AB \overset{2}{\Rightarrow} aaAB \overset{3}{\Rightarrow} aaB \overset{4}{\Rightarrow} aaBb \overset{5}{\Rightarrow} aab$$

Rightmost derivation of $aab$ :

$$S \overset{1}{\Rightarrow} AB \overset{4}{\Rightarrow} ABb \overset{5}{\Rightarrow} Ab \overset{2}{\Rightarrow} aaAb \overset{3}{\Rightarrow} aab$$

# Derivation Trees

Consider the same example grammar:

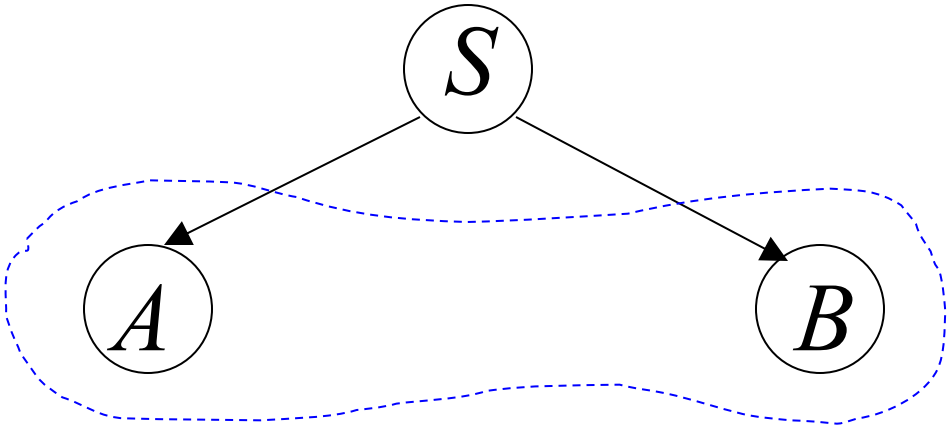
$$S \to AB \qquad A \to aaA \mid \varepsilon \qquad B \to Bb \mid \varepsilon$$

And a derivation of $aab$ :

$$S \Rightarrow AB \Rightarrow aaAB \Rightarrow aaABb \Rightarrow aaBb \Rightarrow aab$$

$S \to AB$ $A \to aaA \mid \varepsilon$ $B \to Bb \mid \varepsilon$

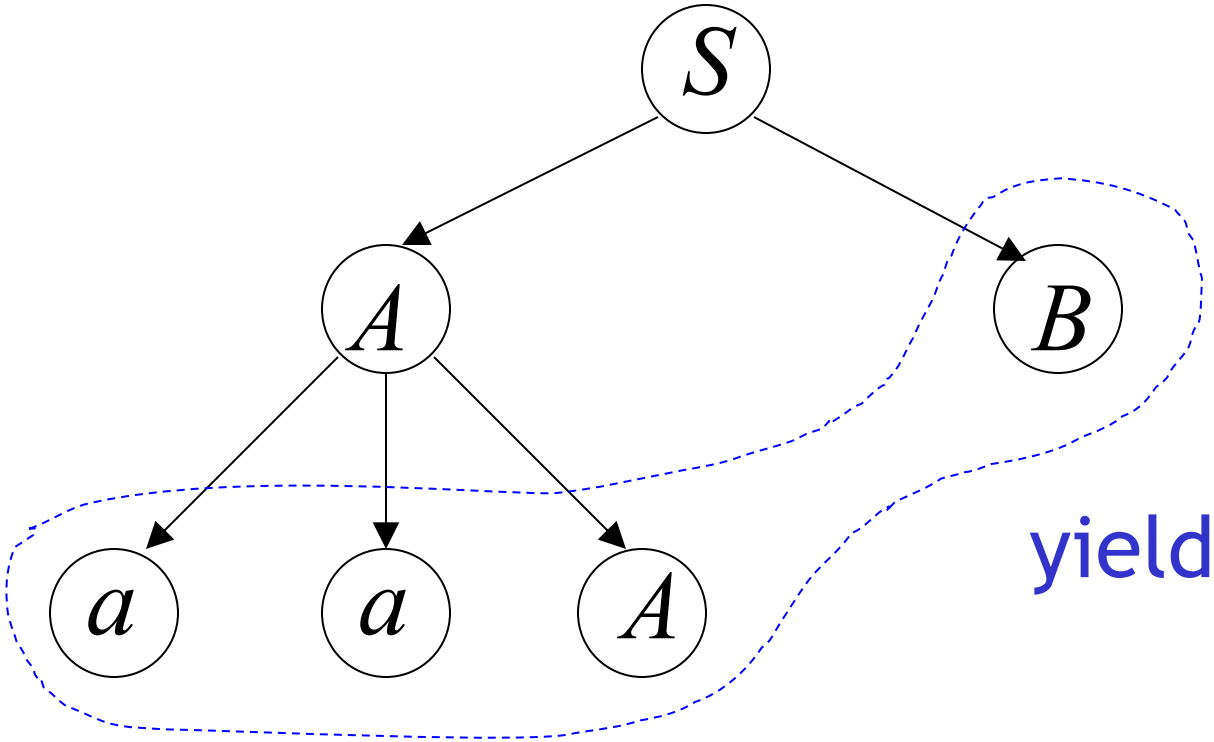
$S \Rightarrow AB$




yield $AB$

$$S \to AB \qquad A \to aaA \mid \varepsilon \qquad B \to Bb \mid \varepsilon$$

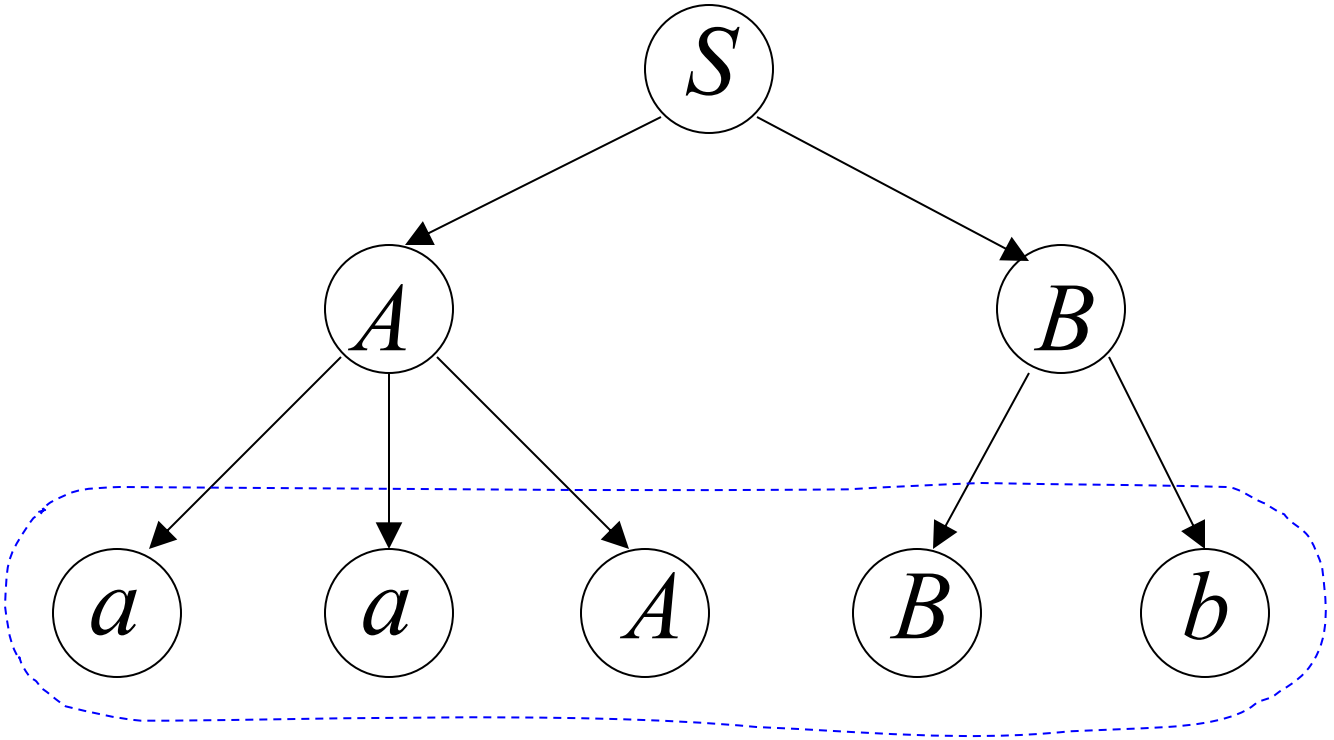
$$S \Rightarrow AB \Rightarrow aaAB$$




yield *aaAB*

$$S \to AB \qquad A \to aaA \mid \varepsilon \qquad B \to Bb \mid \varepsilon$$

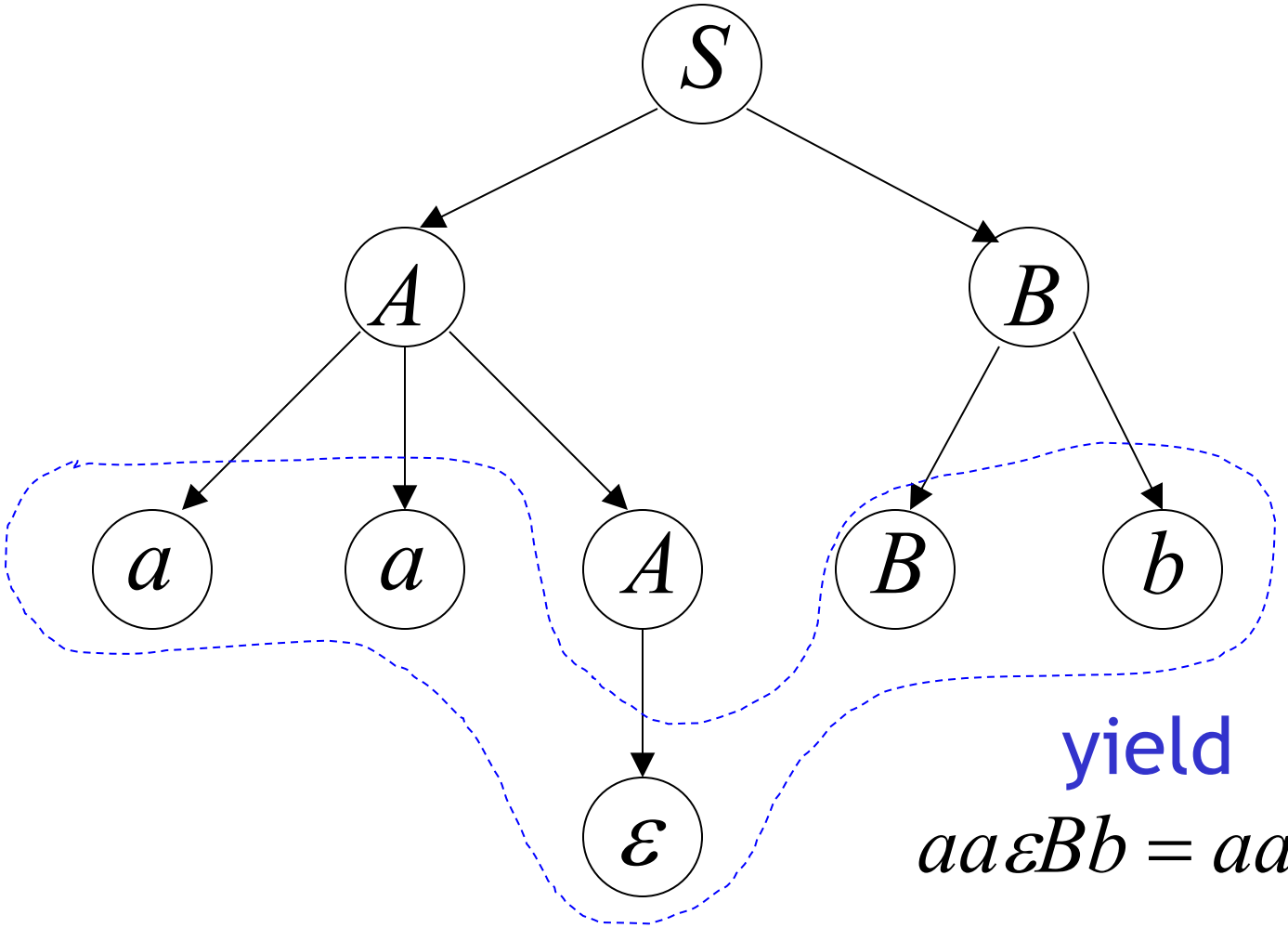
$$S \Rightarrow AB \Rightarrow aaAB \Rightarrow aaABb$$




yield *aaABb*

$$S \to AB \qquad A \to aaA \mid \varepsilon \qquad B \to Bb \mid \varepsilon$$

$$S \Rightarrow AB \Rightarrow aaAB \Rightarrow aaABb \Rightarrow aaBb$$



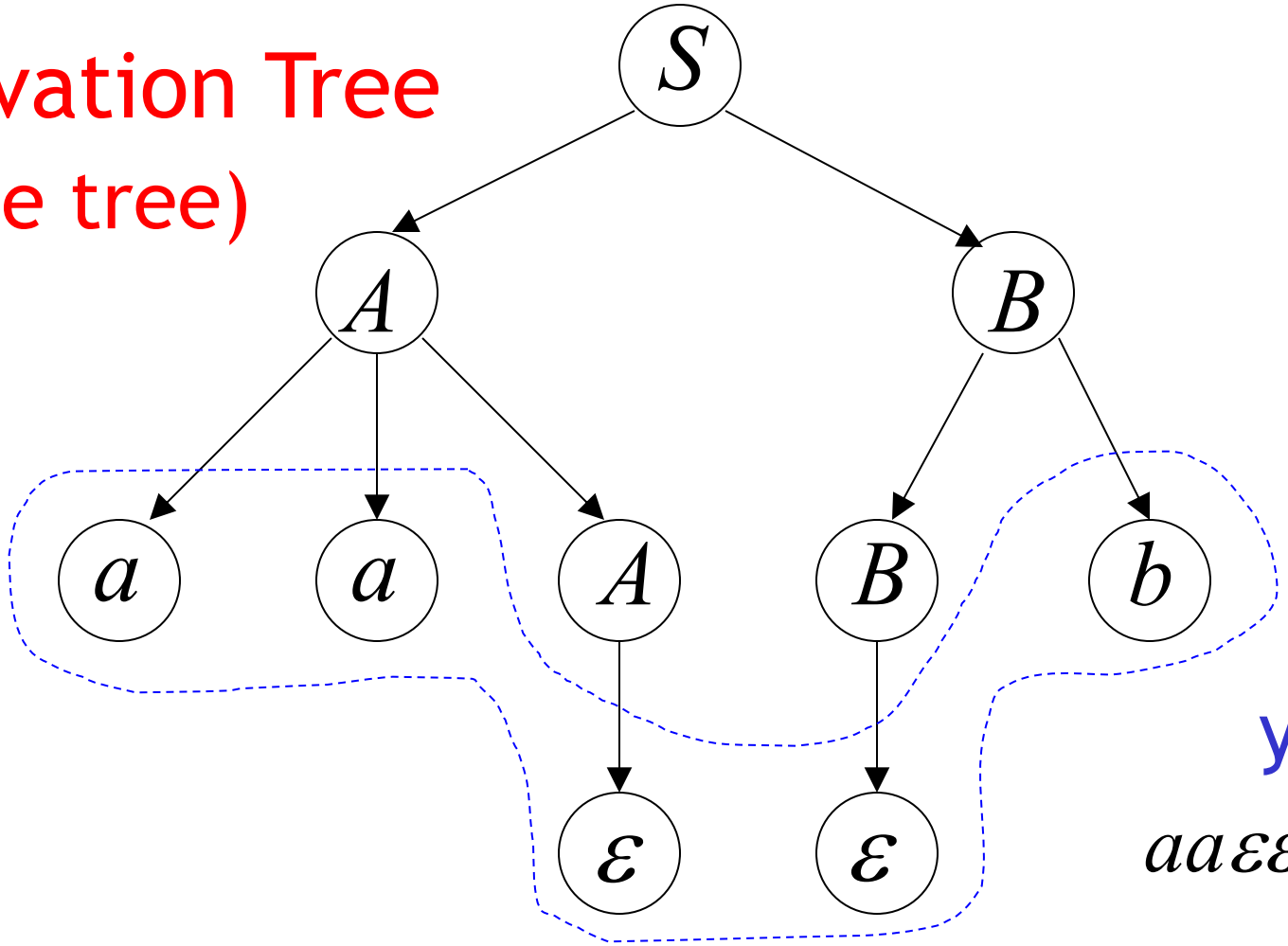

yield

$aa\varepsilon Bb = aaBb$

$$S \to AB \qquad A \to aaA \mid \varepsilon \qquad B \to Bb \mid \varepsilon$$

$$S \Rightarrow AB \Rightarrow aaAB \Rightarrow aaABb \Rightarrow aaBb \Rightarrow aab$$

## Derivation Tree
## (parse tree)




yield

$$aa\varepsilon\varepsilon b = aab$$

Sometimes, derivation order doesn't matter:

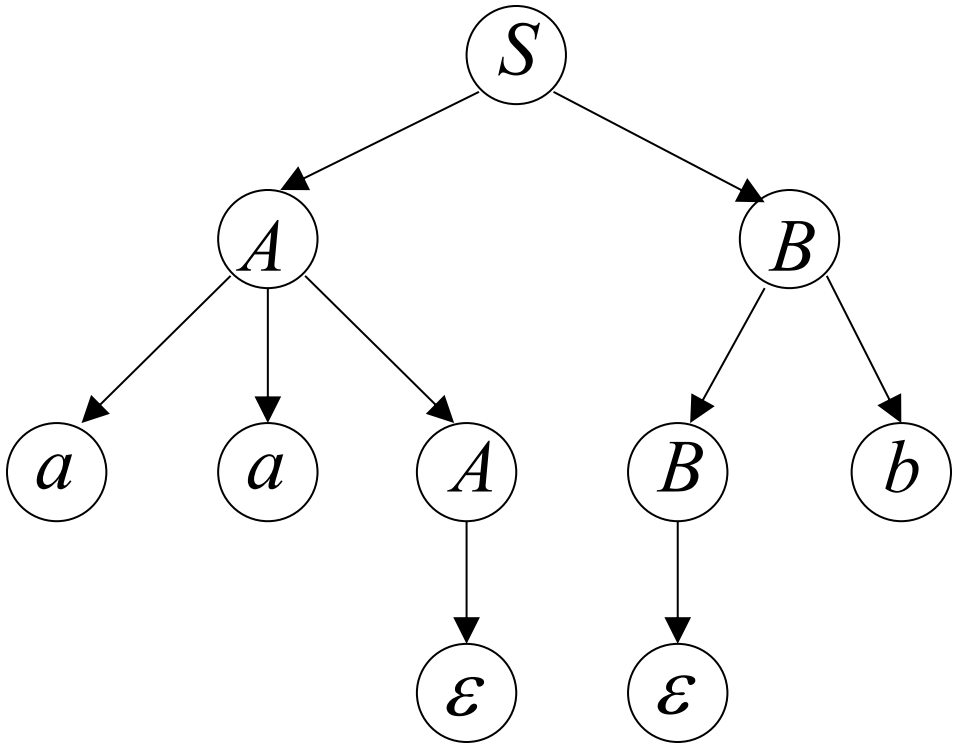
Leftmost derivation:

$$S \Rightarrow AB \Rightarrow aaAB \Rightarrow aaB \Rightarrow aaBb \Rightarrow aab$$

Rightmost derivation:

$$S \Rightarrow AB \Rightarrow ABb \Rightarrow Ab \Rightarrow aaAb \Rightarrow aab$$

Give the same derivation tree



# Ambiguity

# Grammar for mathematical expressions:

$$E \rightarrow E + E \mid E * E \mid (E) \mid a$$

Example strings:

$$(a + a) * a + (a + a * (a + a))$$

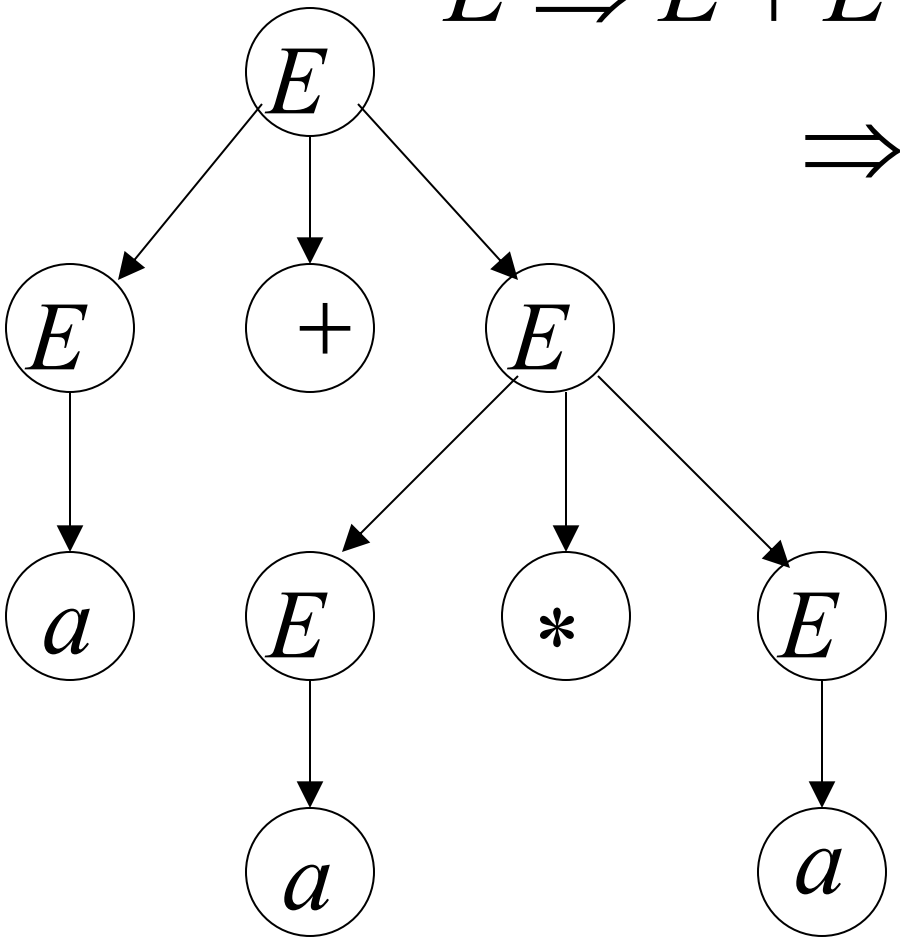


Denotes any number

$$E \rightarrow E + E \mid E * E \mid (E) \mid a$$

$$E \Rightarrow E + E \Rightarrow a + E \Rightarrow a + E * E$$

$$\Rightarrow a + a * E \Rightarrow a + a * a$$



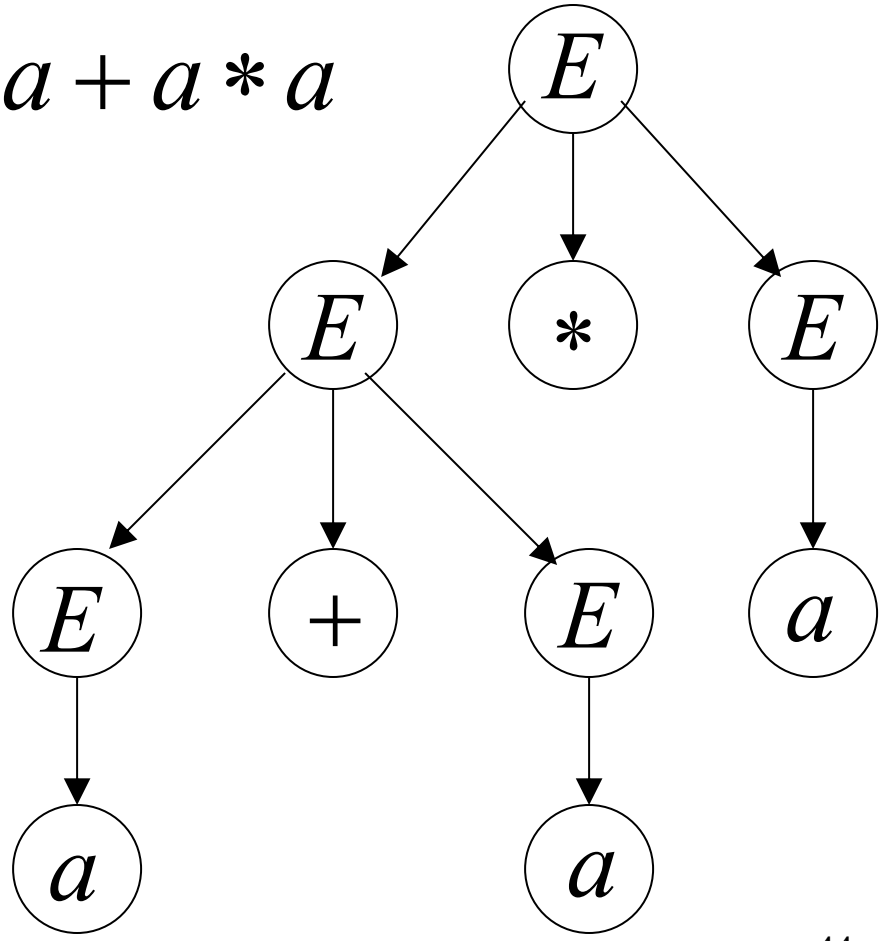

A leftmost derivation for $a + a * a$

$$E \to E + E \mid E * E \mid (E) \mid a$$

$$E \Rightarrow E * E \Rightarrow E + E * E \Rightarrow a + E * E$$

$$\Rightarrow a + a * E \Rightarrow a + a * a$$

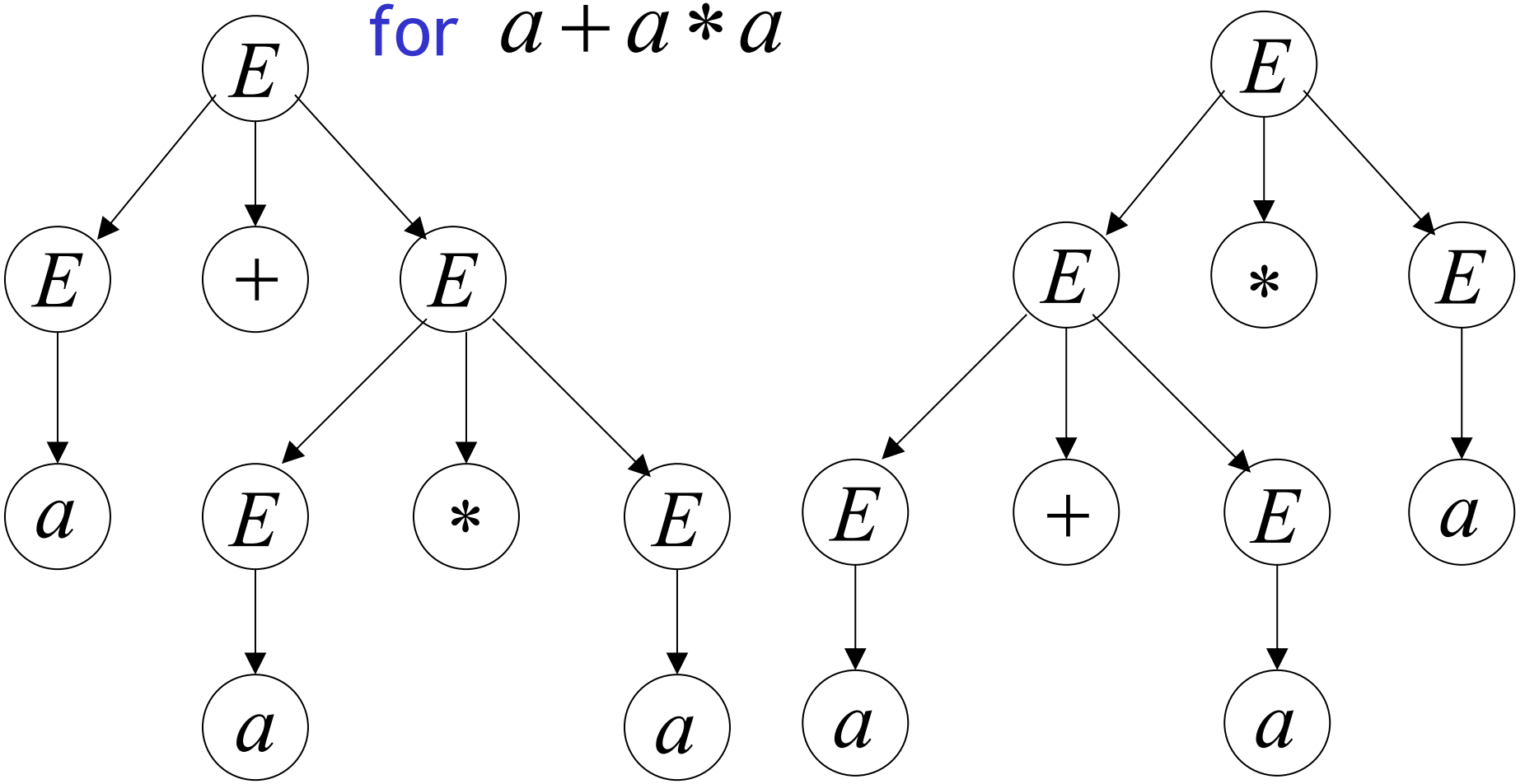
Another leftmost derivation for $a + a * a$



$$E \rightarrow E + E \mid E * E \mid (E) \mid a$$

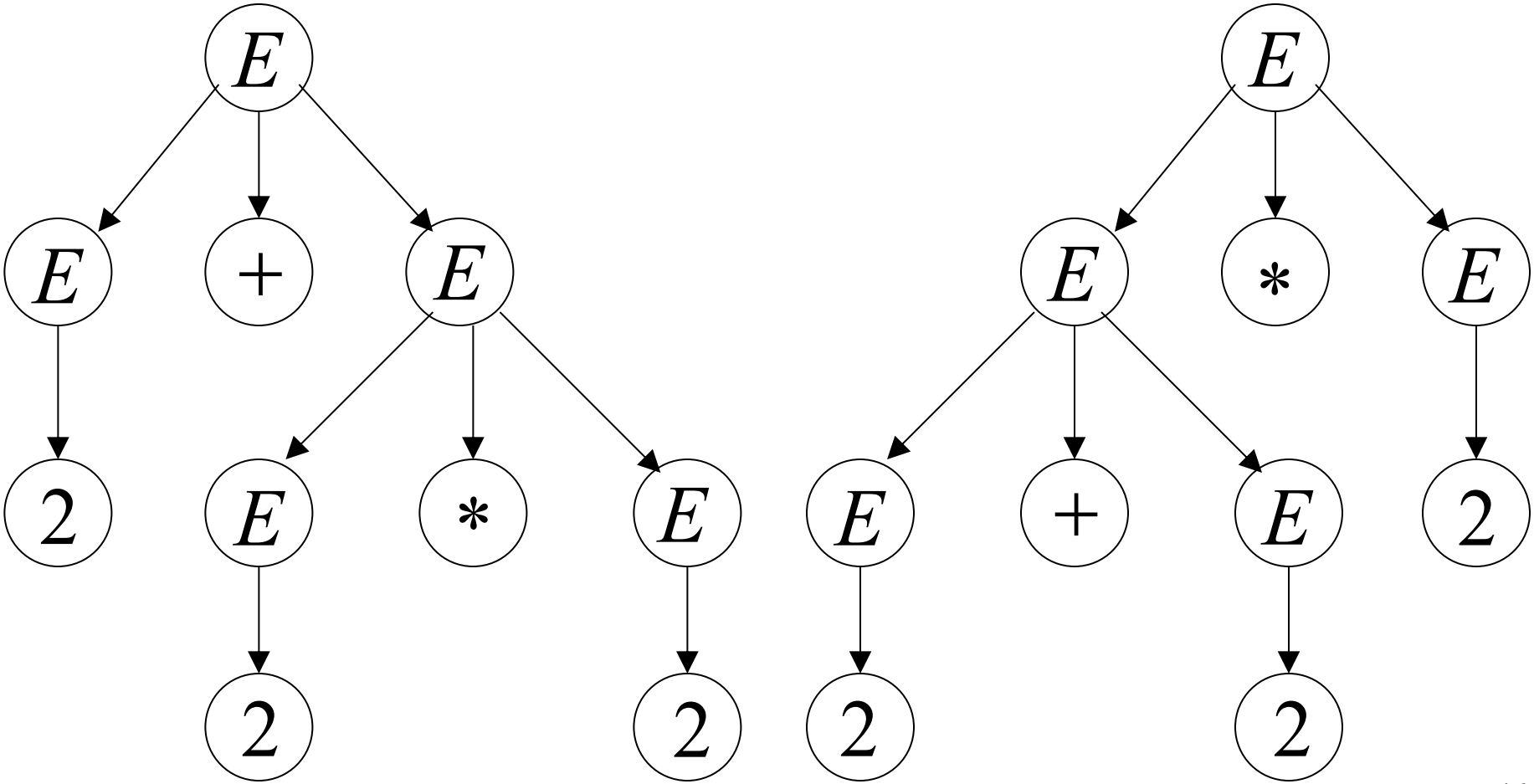
Two derivation trees

for $a + a * a$



take $a = 2$

$$a + a * a = 2 + 2 * 2$$



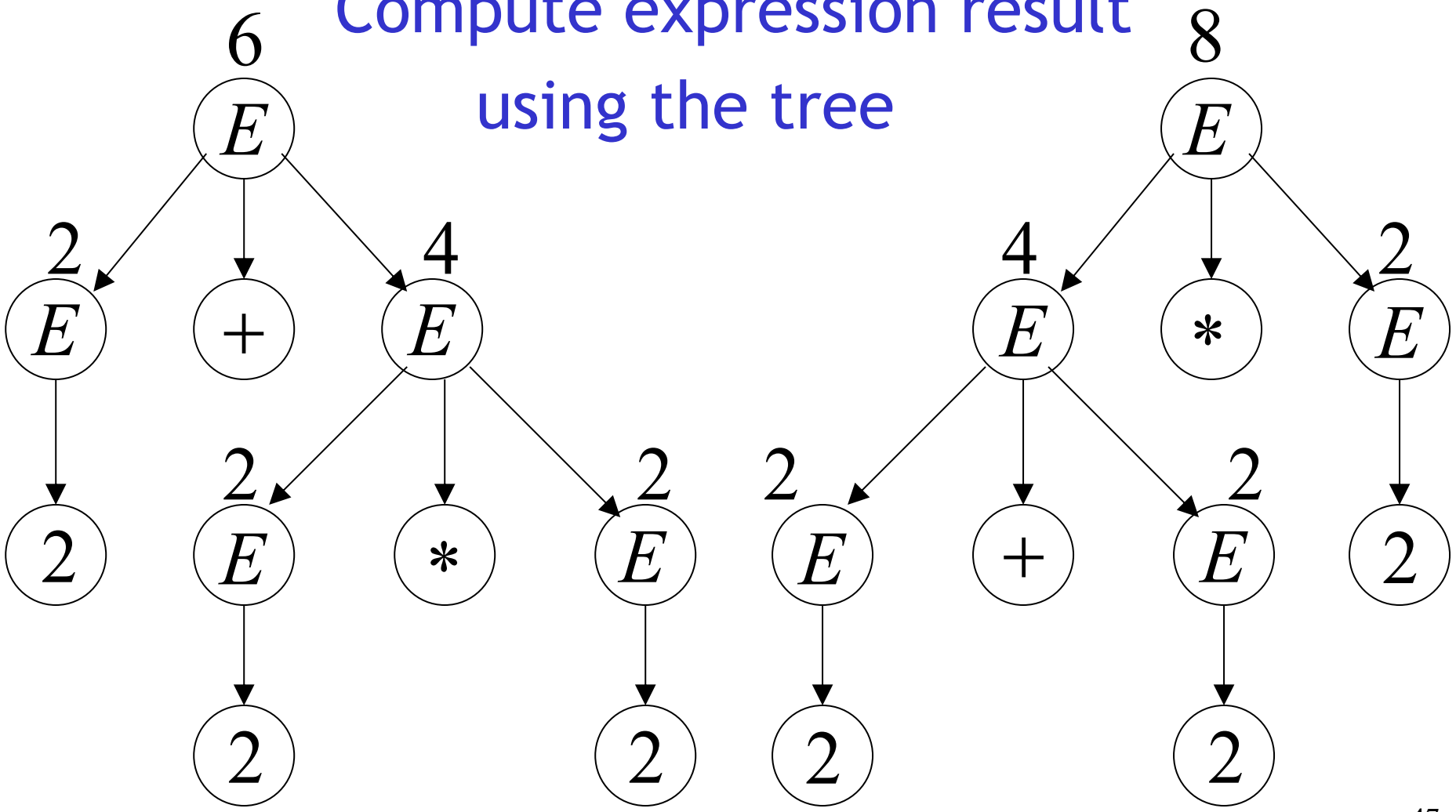
Good Tree

$2+2*2=6$

Bad Tree

$2+2*2=8$

Compute expression result using the tree



Two different derivation trees
may cause problems in applications which use the derivation trees:

- Evaluating expressions

- In general, in compilers for programming languages

# Ambiguous Grammar:

A context-free grammar $G$ is ambiguous if there is a string $w \in L(G)$ which has:

two different derivation trees

or

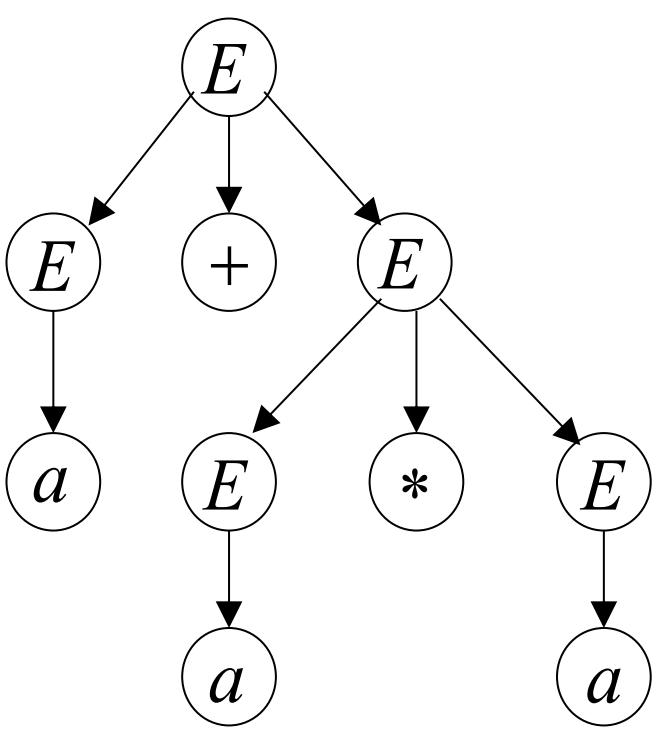
two leftmost derivations

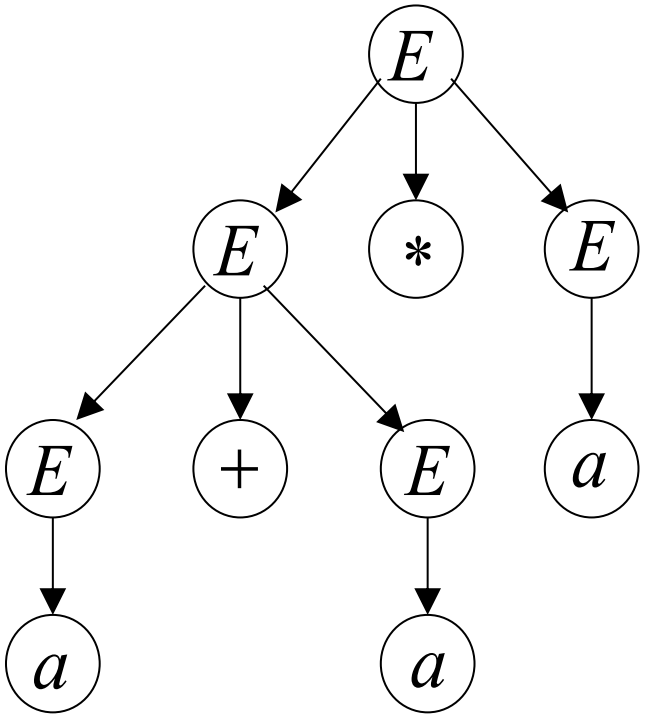
(Two different derivation trees give two different leftmost derivations and vice-versa.)

Example: $E \rightarrow E+E \mid E*E \mid (E) \mid a$

This grammar is ambiguous since

The string $a+a*a$ has two derivation trees:



$$E \to E + E \mid E * E \mid (E) \mid a$$

This grammar is ambiguous also because the string $a + a * a$ has two leftmost derivations:

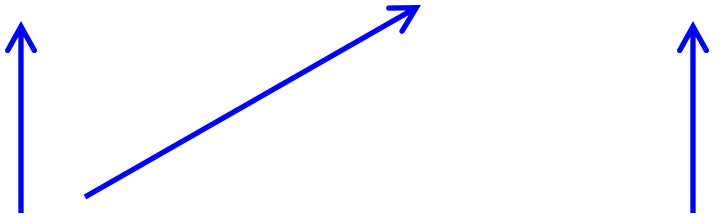
$$E \Rightarrow E + E \Rightarrow a + E \Rightarrow a + E * E$$
$$\Rightarrow a + a * E \Rightarrow a + a * a$$

$$E \Rightarrow E * E \Rightarrow E + E * E \Rightarrow a + E * E$$
$$\Rightarrow a + a * E \Rightarrow a + a * a$$

# Another ambiguous grammar:

STMT → if EXPR then STMT

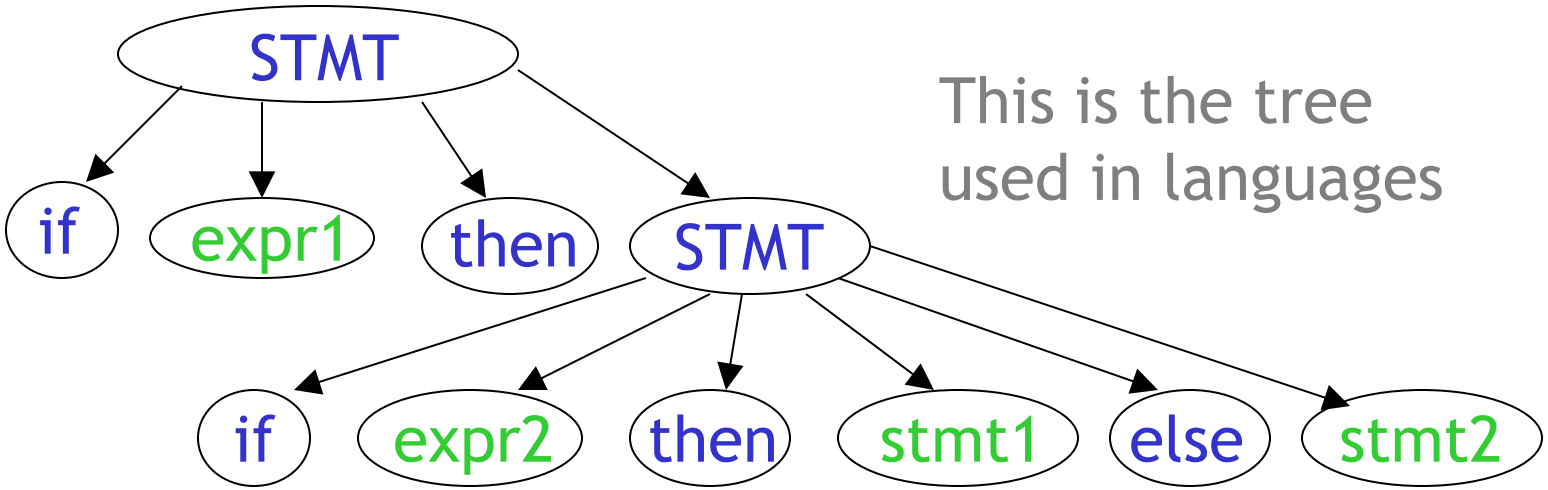
| if EXPR then STMT else STMT



Variables Terminals

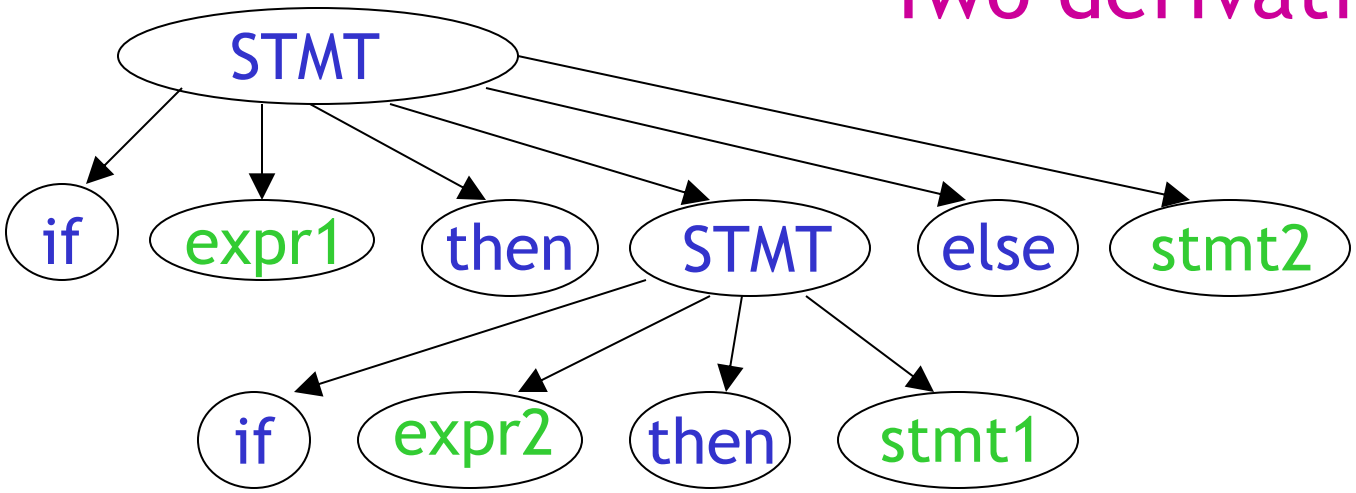
Very common piece of grammar in programming languages.

# If expr1 then if expr2 then stmt1 else stmt2




This is the tree used in languages

## Two derivation trees



In general, ambiguity is bad
and we want to remove it

Sometimes it is possible to find
a non-ambiguous grammar for a language.

But, in general it is difficult to achieve this.

# A successful example:

Ambiguous Grammar

$$E \to E + E$$
$$E \to E * E$$
$$E \to (E)$$
$$E \to a$$

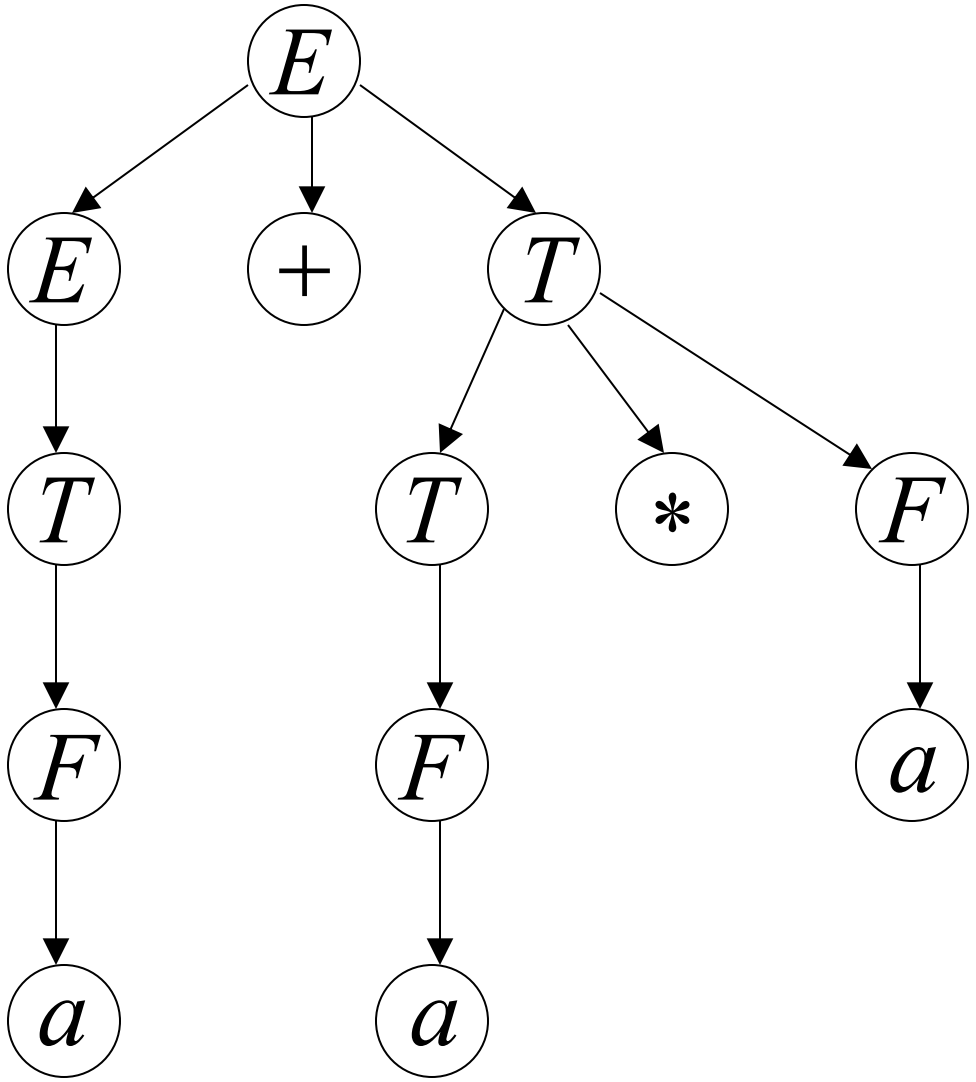
Equivalent Non-Ambiguous Grammar

$$E \to E + T \mid T$$
$$T \to T * F \mid F$$
$$F \to (E) \mid a$$

generates the same language

$$E \Rightarrow E+T \Rightarrow T+T \Rightarrow F+T \Rightarrow a+T \Rightarrow a+T*F$$
$$\Rightarrow a+F*F \Rightarrow a+a*F \Rightarrow a+a*a$$

$$E \rightarrow E+T \mid T$$
$$T \rightarrow T*F \mid F$$
$$F \rightarrow (E) \mid a$$




Unique derivation tree for $a+a*a$

## An un-successful example:

$$L = \{a^n b^n c^m\} \cup \{a^n b^m c^m\}$$

$$n, m \geq 0$$

$L$ is inherently ambiguous:

Every grammar that generates this language is ambiguous.

# Example (ambiguous) grammar for $L$ :

$$L = \{a^n b^n c^m\} \cup \{a^n b^m c^m\}$$

$$S \to S_1 \mid S_2 \qquad S_1 \to S_1 c \mid A \qquad S_2 \to a S_2 \mid B$$

$$A \to aAb \mid \varepsilon \qquad B \to bBc \mid \varepsilon$$

The string $a^n b^n c^n \in L$

has always two different derivation trees

(for any grammar)

For example

